

Short-Term Wind Energy Forecasting in Ireland with LSTM, SARIMA, and Random Forest

Context & Motivation

Ireland's goal is to generate 80% of its electricity from renewable sources by 2030. However, wind energy is volatile and poses a challenge to grid stability.

An accurate short-term forecast (three to seven days) would reduce reliance on fossil fuels, optimise storage and improve planning.

Data:

Wind power generation in Ireland: EirGrid, 15 min $\rightarrow$ daily aggregation (2013–2025). Given the interest in forecasting daily wind power generation levels, the time series has been resampled to a daily frequency.

Hypothesis:

LSTM networks will achieve higher short-term wind generation forecasting accuracy than Random Forest and SARIMA, due to their ability to learn non-linear temporal dependencies in sequential data.

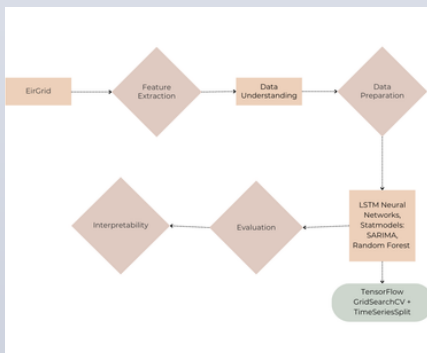
General goal:

To develop and compare three forecasting models—LSTM, SARIMA, and Random Forest—using historical national wind-generation data, in order to identify which approach provides the most accurate short-term forecasts for the Irish grid.

Success Criteria:

- Lower RMSE across all models
- Low MAE as an indicator of average prediction error
- Consistent performance across multiple forecast horizons (3, 5, 7 days)
- Model interpretability supported by SHAPASH and LIME

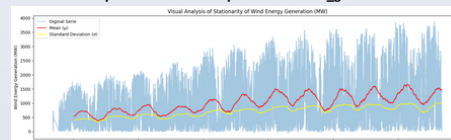
Methodology (CRISP-DM)



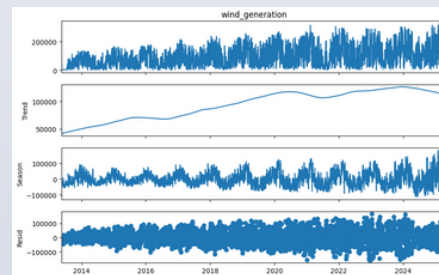
Models

Model	Key Feature	Forecast Horizon	Notes
SARIMA	Seasonal ARIMA (7 day seasonality)	3, 5, 7 days	Interpretable, captures autocorrelation well.
Random Forest	Lag features, flattened windows	3, 5, 7 days	Uses lag features as predictors Must train one model per horizon step Very time-efficient Learns non-linear patterns
LSTM	Sequential input, multistep output.	3, 5, 7 days	Learns temporal patterns and non-linear relationships. Uses memory cells with three gates (forget, input, output) to retain long-term information

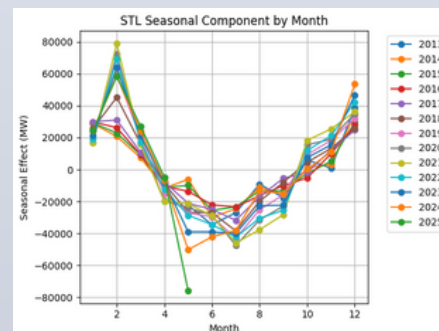
Representation of the pattern of 'wind_generation'



STL Decomposition



STL Seasonal Plot Average Wind Generation by Month (per Year)



Evaluation Framework

Training procedure

The series was divided into segments:

- 80% training
- 10% validation
- 10% test

Evaluated configurations

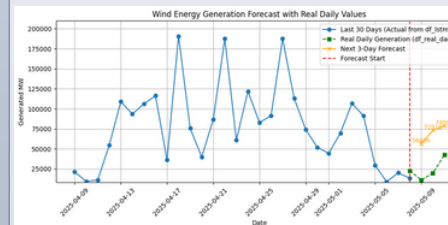
Input history: 10, 20, 30 days
Forecast horizons: 3, 5, 7 days

Hyperparameters

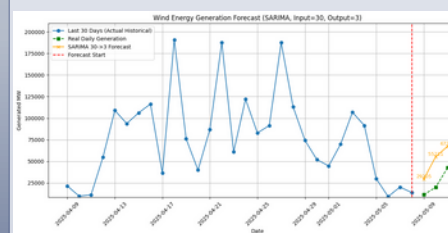
They were tuned using GridSearchCV with a 5-fold TimeSeriesSplit to ensure chronological validation, and each LSTM model was automatically tuned using Keras Tuner (Random Search).

Wind Energy Generation Forecast

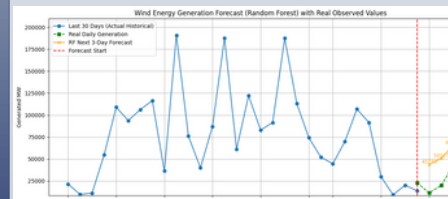
LSTM



SARIMA



RANDOM FOREST



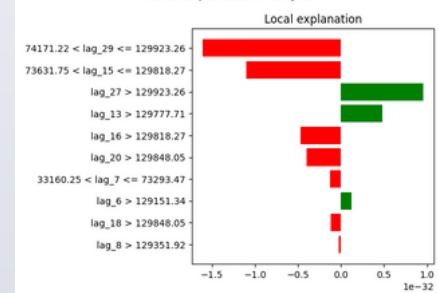
Results

Comparison of Best Configurations cross Models (RMSE and MAE)

Model	Best Configuration (Input - Output)	RMSE	MAE
SARIMA	10 --- 3	76,270	69,994
Random Forest	30 --- 3	69,341	56,195
LSTM	20 --- 3	69,448	56,271

Model Interpretability

LIME Explanation - Day 3



Shapash



Conclusion

Random Forest achieved the best forecasting accuracy, outperforming both SARIMA and LSTM for short-term wind generation prediction.

SARIMA worked well as a baseline model, capturing weekly patterns despite the variability of the data.

LSTM required much more computation and did not provide better accuracy under the tested settings.

In summary, Random Forest was the most effective model, although it needs one model per forecast step, while LSTM can produce multi-step forecasts in a single model.